1 A student is investigating how the intensity of the reflection of sound from a wall varies with the thickness of foam attached to the wall, as shown in Fig. 1.1.

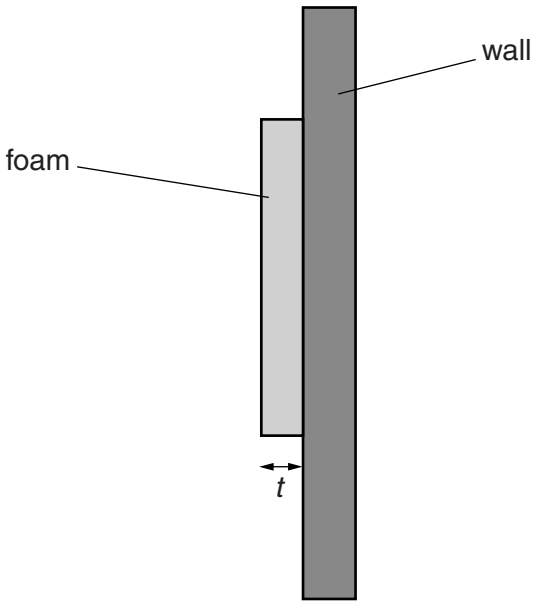

Fig. 1.1

It is suggested that the intensity I of the reflected sound is related to the thickness t of the foam by the relationship

$$I = I_0 e^{-\alpha\rho t}$$

where I_0 is the intensity of the sound before reflection, ρ is the density of the foam and α is a constant.

Design a laboratory experiment to test the relationship between I and t. Explain how your results could be used to determine a value for α. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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